

Navigation

Navigation is the art and science of finding a way from one place to another, along with controlling and monitoring.

Primary Instruments

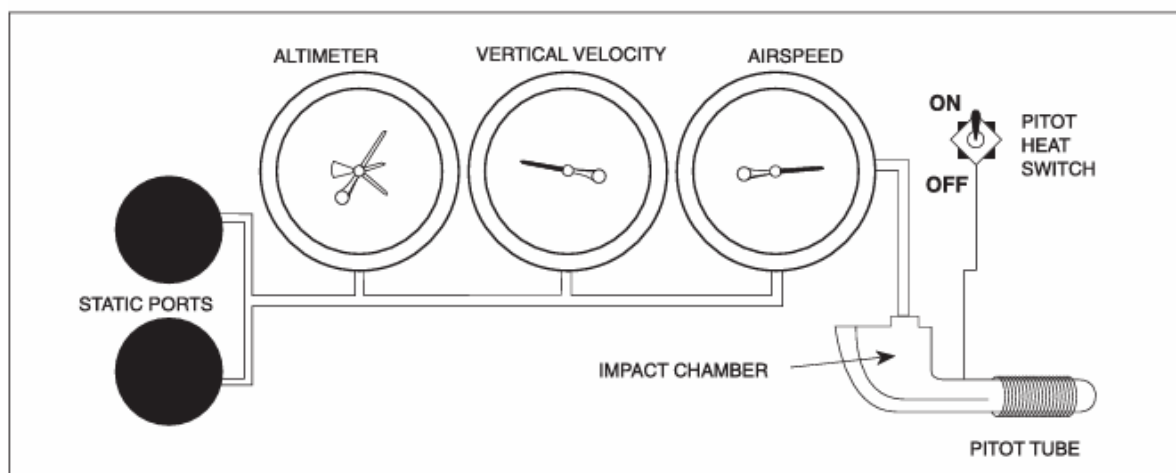
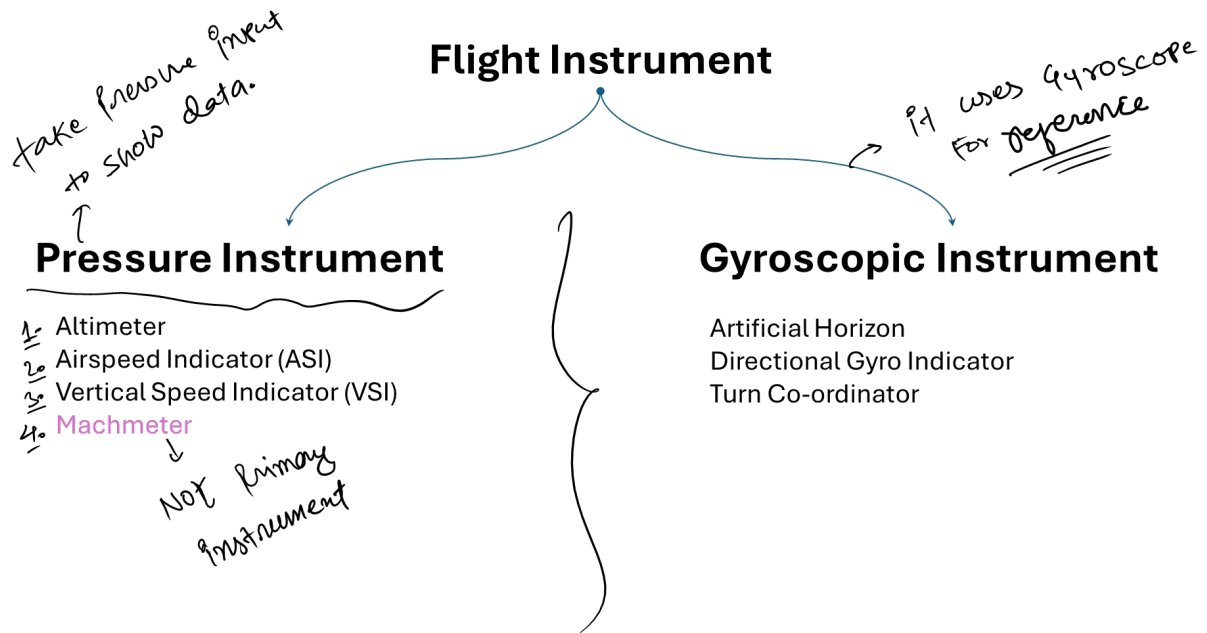
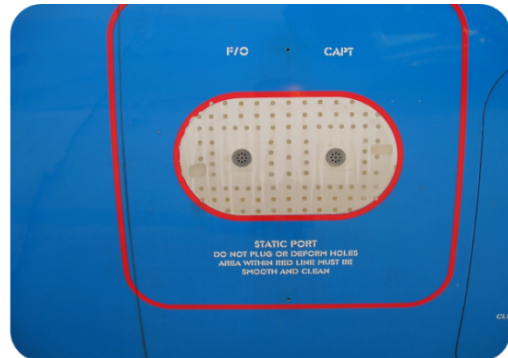


Figure 3-1. Pitot-static system



Pitot Tube

it senses $P_s + P_d$



Static Port

it senses P_s only

Pitot-Static Instruments

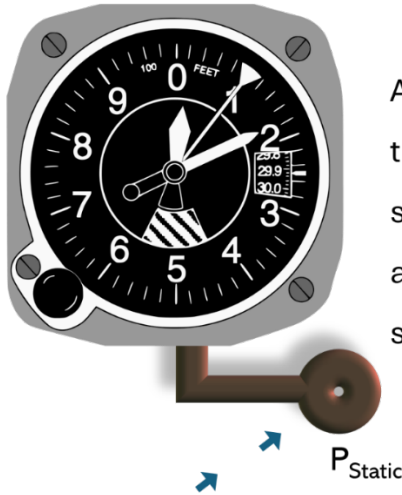
The pressure altimeter, vertical-speed indicator, and airspeed indicator operate in response to pressures through the pitot-static system. See Figure 3-1.

Static (atmospheric) pressure is taken from the static vents and is provided to all three instruments. *Clogging of the static vents or line will cause all three instruments to become inoperative or to display erroneous readings.*

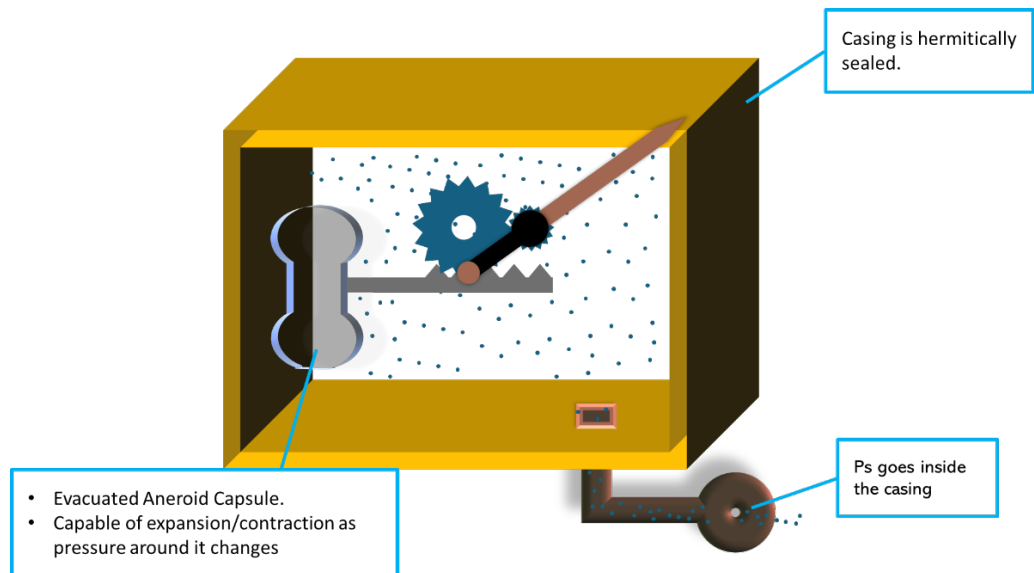
Ram/pitot pressure is taken from the pitot tube and furnished to the airspeed indicator only. *Clogging of the pitot opening will not affect operation of the altimeter or vertical speed indicator.*

Altimeter

Principal of Altimeter



An Altimeter contains aneroid barometer that takes input from static port and by measuring static pressure shows the corresponding altitude According to ISA , w.r.t subscales setting.



A/c is descending
↓
Ps ↑↑↑
↓
Case Pressure ↑↑↑
↓
Capsule compresses



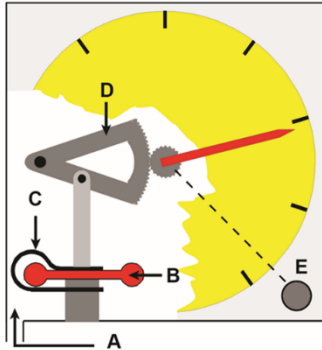
A/c is climbing
↓
Ps ↓↓↓↓
↓
Case Pressure ↓↓↓
↓
Capsule expands



Type of Altimeter

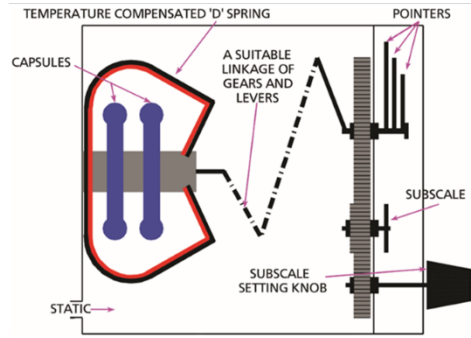
1

Simple Altimeter



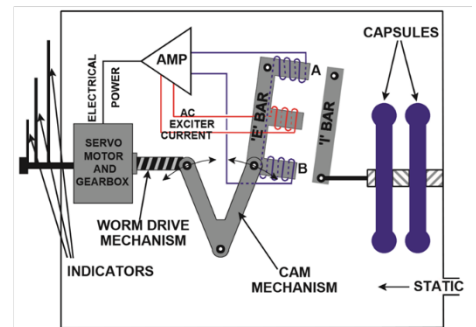
2

Sensitive Altimeter



3

Servo-assisted Altimeter



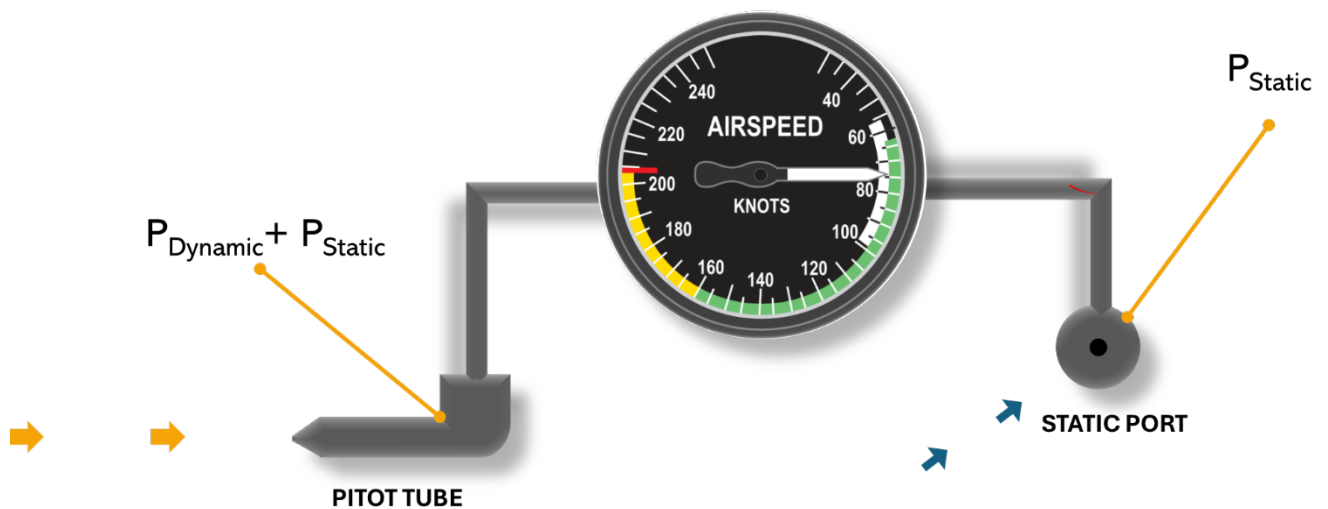
Radio Altimeter

Radio altimeter is an instrument fitted in the aircraft. It uses primary radar principle and shows height above the ground in feet.

It uses frequency modulated carrier waves of frequency 4200 MHz to 4400MHz.

Range of radio altimeter is 0 to 2500 feet.

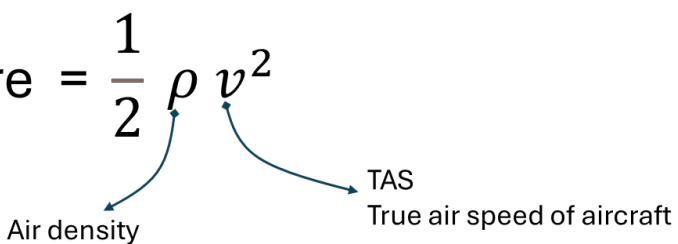
Air Speed Indicator



An ASI consist of differential pressure gauge that measures the dynamic pressure by subtracting static from pitot pressure(total Pressure) and shows airspeed in knots

$$\text{Total Pressure} - \text{Static Pressure} = \text{Dynamic Pressure}$$

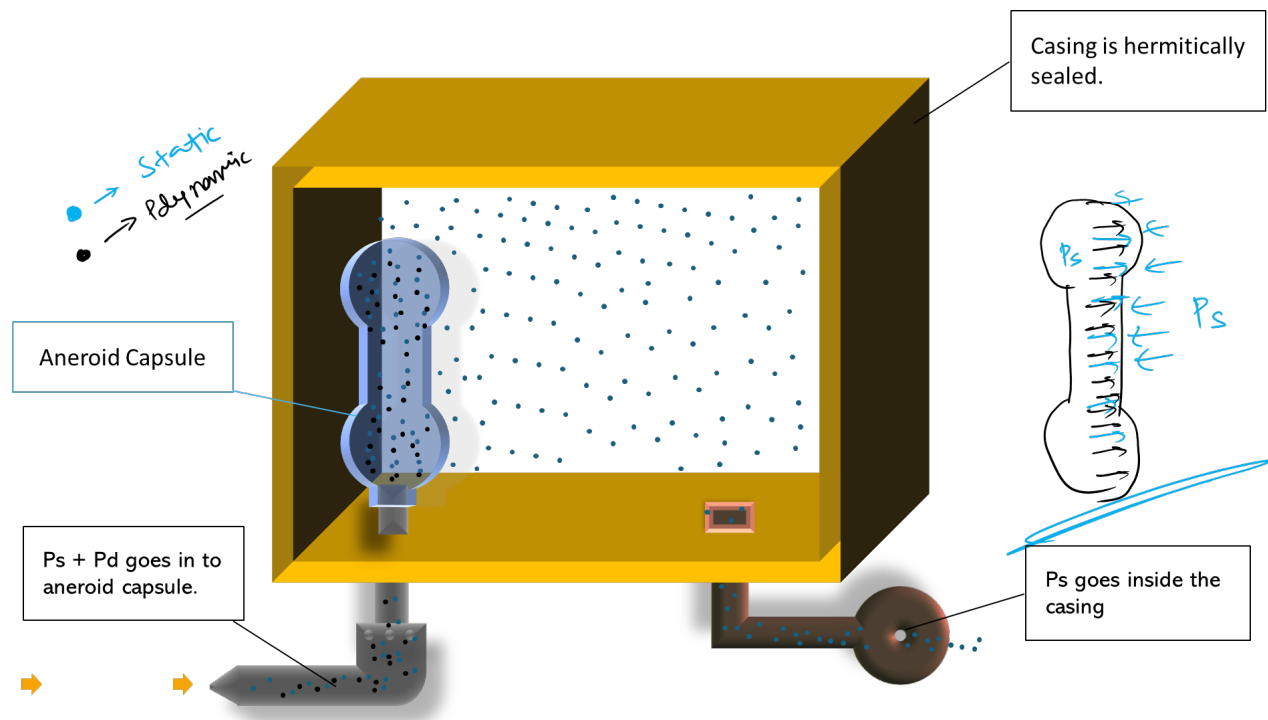
As we know dynamic pressure is the measure of speed.

$$\text{Dynamic Pressure} = \frac{1}{2} \rho v^2$$


The diagram shows the formula $\text{Dynamic Pressure} = \frac{1}{2} \rho v^2$. Below the formula, there are two labels with arrows pointing to the variables: 'Air density' points to the Greek letter ρ , and 'TAS True air speed of aircraft' points to the variable v .

So, by measuring Dynamic pressure we can get Airspeed of aircraft.

ASI Working



Dynamic pressure cannot be directly measured; it must be established from the difference between pitot and static pressure.

Static pressure is fed into a hermetically-sealed instrument case, the pitot pressure being fed to a thin metal capsule capable of expansion and contraction.

Static pressure is present on both side (inside and the outside) the metal walls of the capsule and so get cancelled. $(D_y + S) - S$ remaining pressure inside the capsule is Dynamic only.

Hence, Expansion or contraction of the capsule will be proportional to the changes in dynamic pressure produced by changes of airspeed.

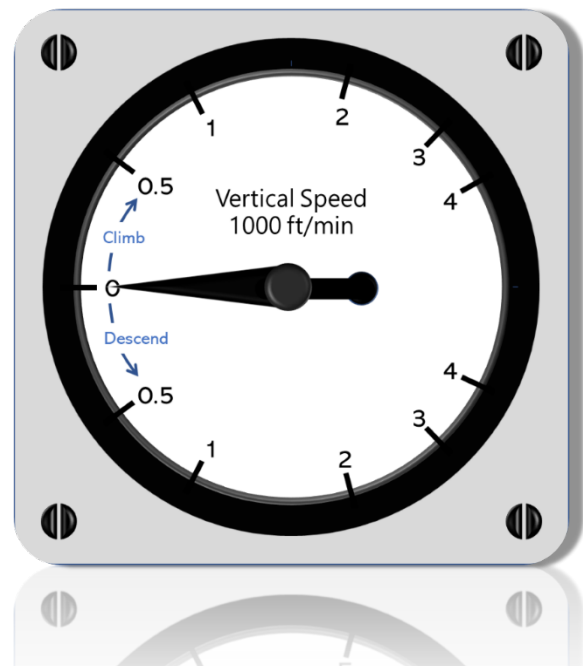
The capsule is connected with a temperature-compensated linkage which moves the pointer and shows airspeed on dial.

VSI

VSI

Vertical Speed Indicator

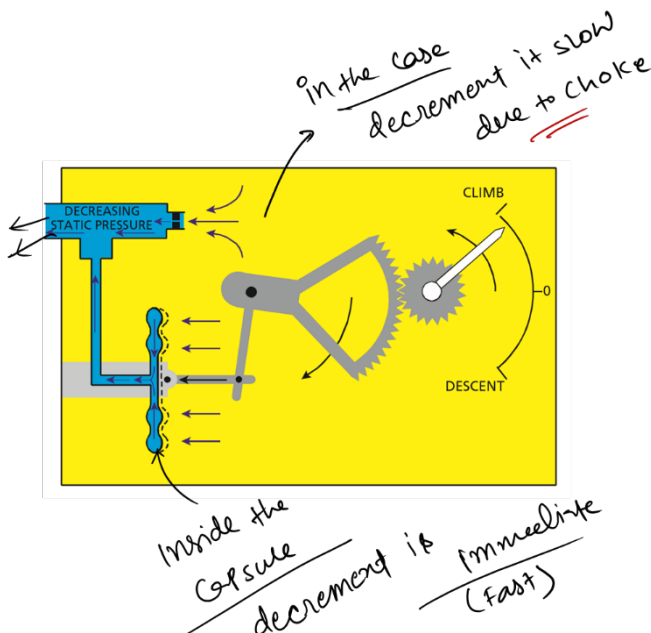
We need VSI while achieving a certain altitude within a specified time, to maintain smooth rate of descent on a glide path on an instrument approach.



How VSI Works

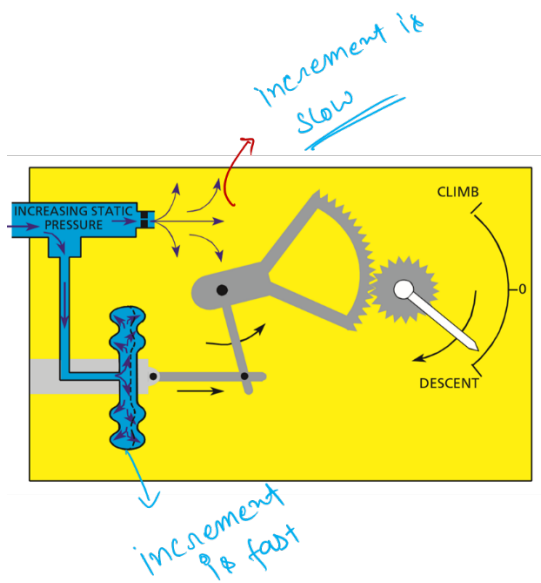
In level flight, the pressure inside the capsule and the case will be the same, so the pointer will remain in its horizontal position, indicating a zero rate of climb.

If the aeroplane climbs, the static pressure in the capsule will decrease at a quicker rate than that in the casing and the capsule will compress slightly and will cause the pointer to indicate a rate of climb.



When Air climbs
 $P_{s \text{ falls}}$
 $\downarrow$
 In the case P_s is higher
 $\downarrow$
 Capsule compresses
 $\downarrow$
Show Climb

Conversely, if the aeroplane descends the static pressure in the capsule will increase at a higher rate than that in the casing and the capsule will expand slightly, causing the pointer to indicate a rate of descent.



descent

P_{s111}

↓

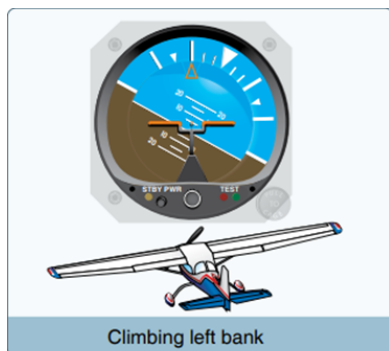
Capsule has higher pressure

↓

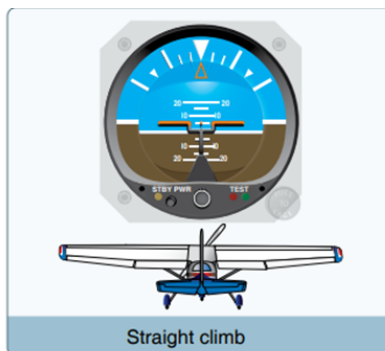
expand → shows descent



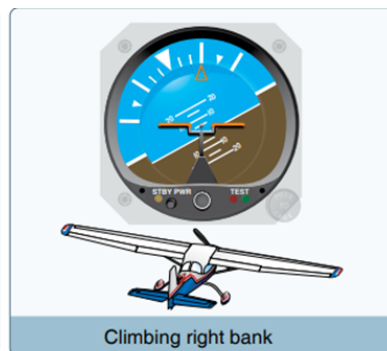
Reference Axis definitions



Climbing left bank



Straight climb



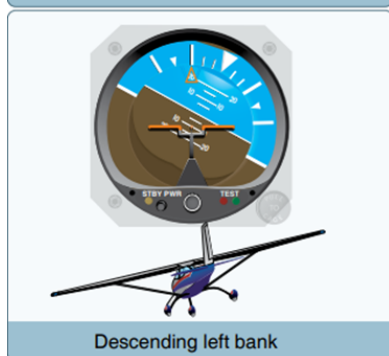
Climbing right bank



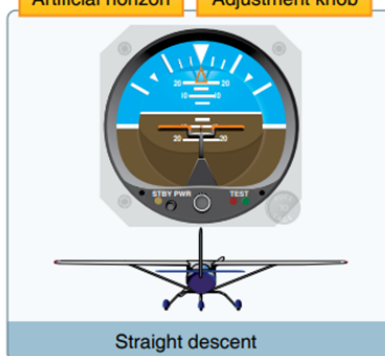
Level left bank



Level right bank



Descending left bank

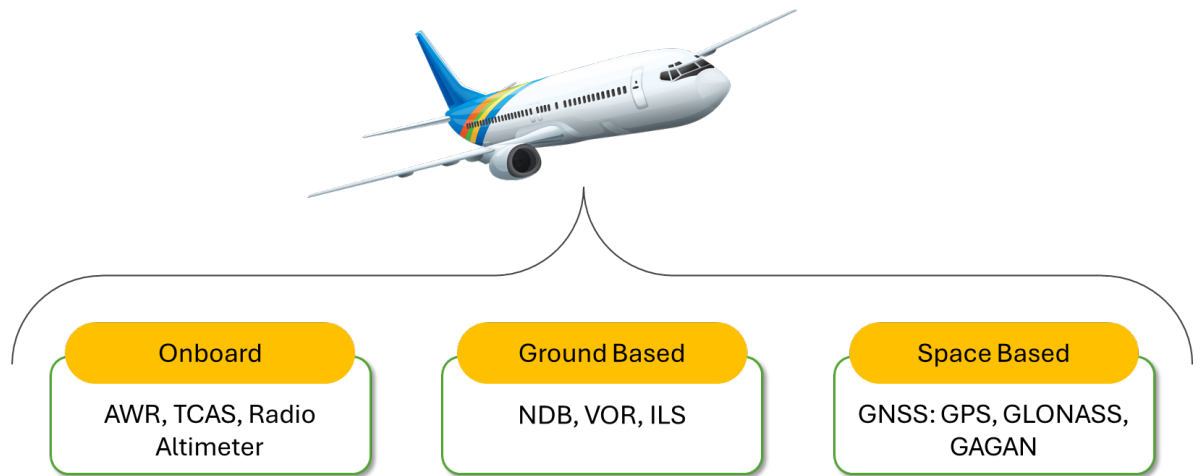


Straight descent



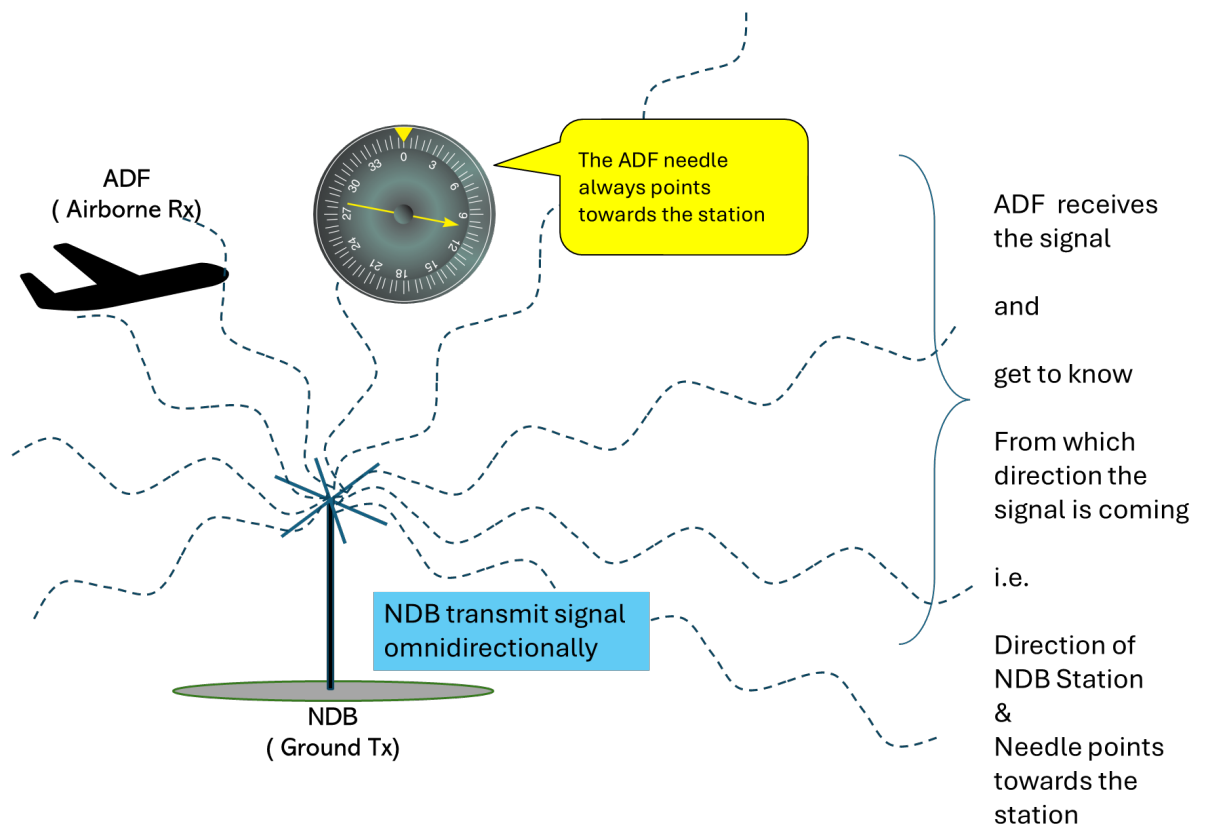
Descending right bank

Radio Navigation



ADF & NDB

- NDB always works in collaboration with ADF.
- Non-Directional Beacons (NDB) are ground-based transmitters that transmit radio signal equally in all directions.
- The airborne system in the aircraft is the Automatic Direction Finder (ADF) which receives the signal.



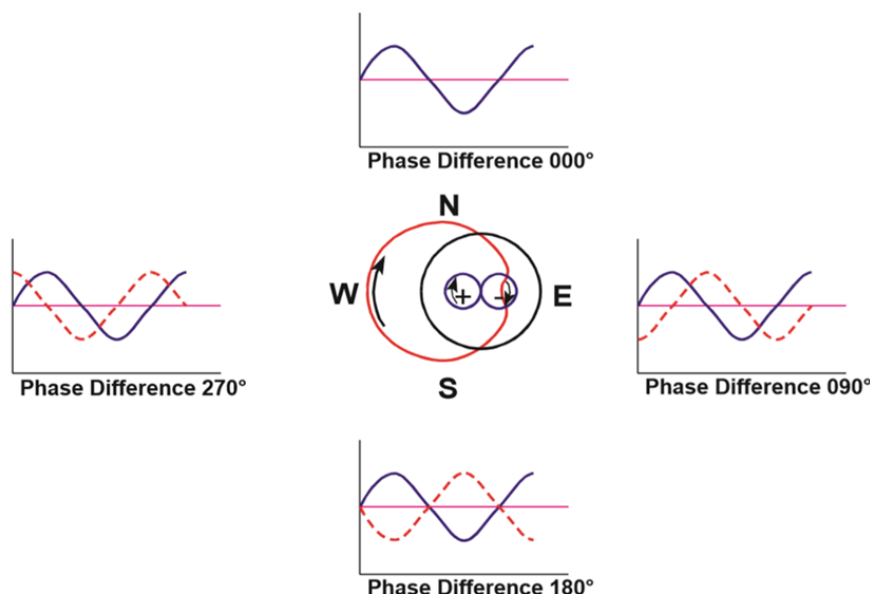
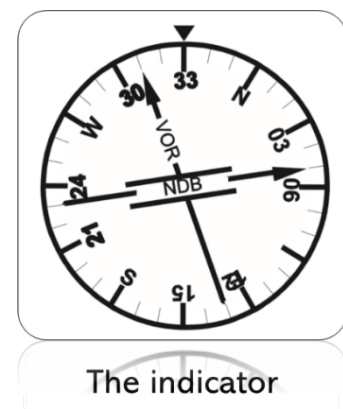
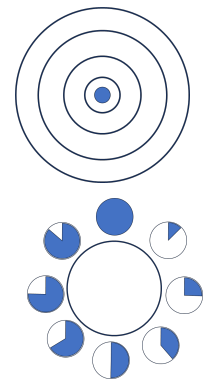
- After receiving the signal, it got to know from which direction the signal is coming and accordingly the needle points towards the station.
- then by flying in that direction aircraft reaches to the destination.



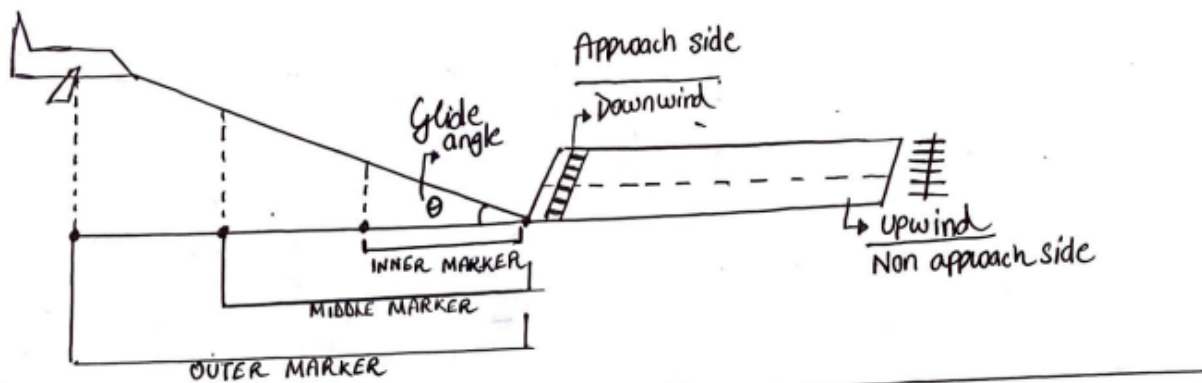
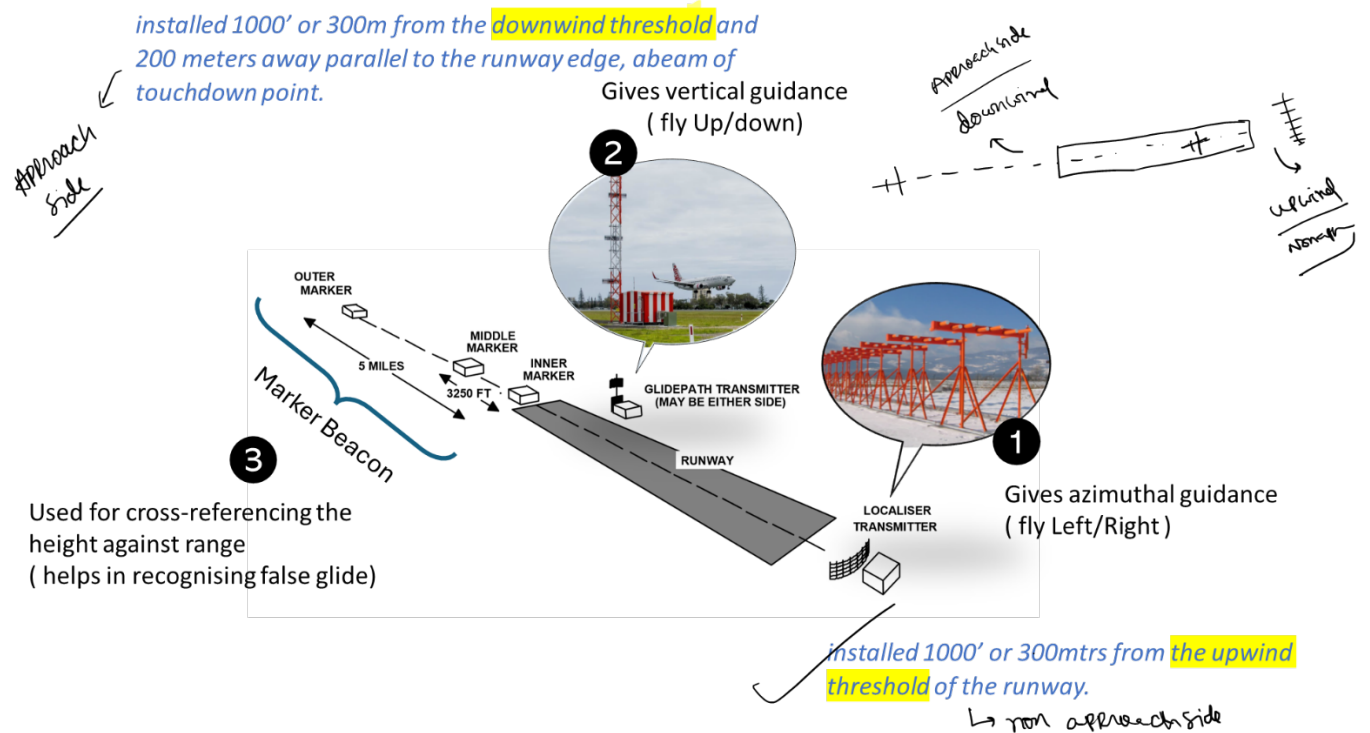
VOR (Very High Frequency Omnidirectional Range)



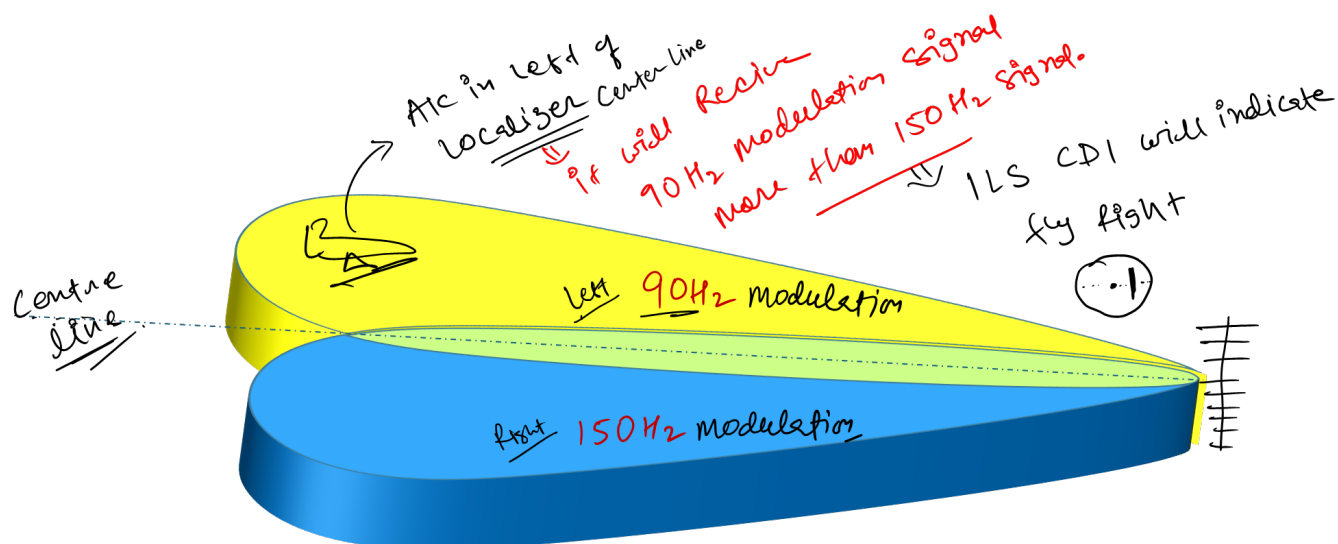
- ▶ VOR ground transmitter has two aerials one is fixed, and another is electronically rotating @ 30 Hz.
- ▶ Fixed aerial transmits a horizontally polarized continuous wave signal of 30 Hz frequency modulation and called reference signal. This signal is received in same phase in all directions. Polar diagram of fixed aerial is a circle.
- ▶ A directional or variable signal is transmitted by a rotating aerial which rotates electronically at 30 Hz or 30 cycles / sec. This signal is AM and received in different phase in all directions.
- ▶ At magnetic North, variable and reference signal are in phase (phase diff zero).
- ▶ The transmission is done in such a way that the **phase difference between two transmissions at any point around VOR will be always equal to the QDR** (i.e. Magnetic bearing of A/C from the station.) QDR.
- ▶ A/C equipment finds the phase difference and gives QDR's or Magnetic Bearing from the station which are known as radial.
- ▶ According to QDR, The RMI needle will point.



ILS

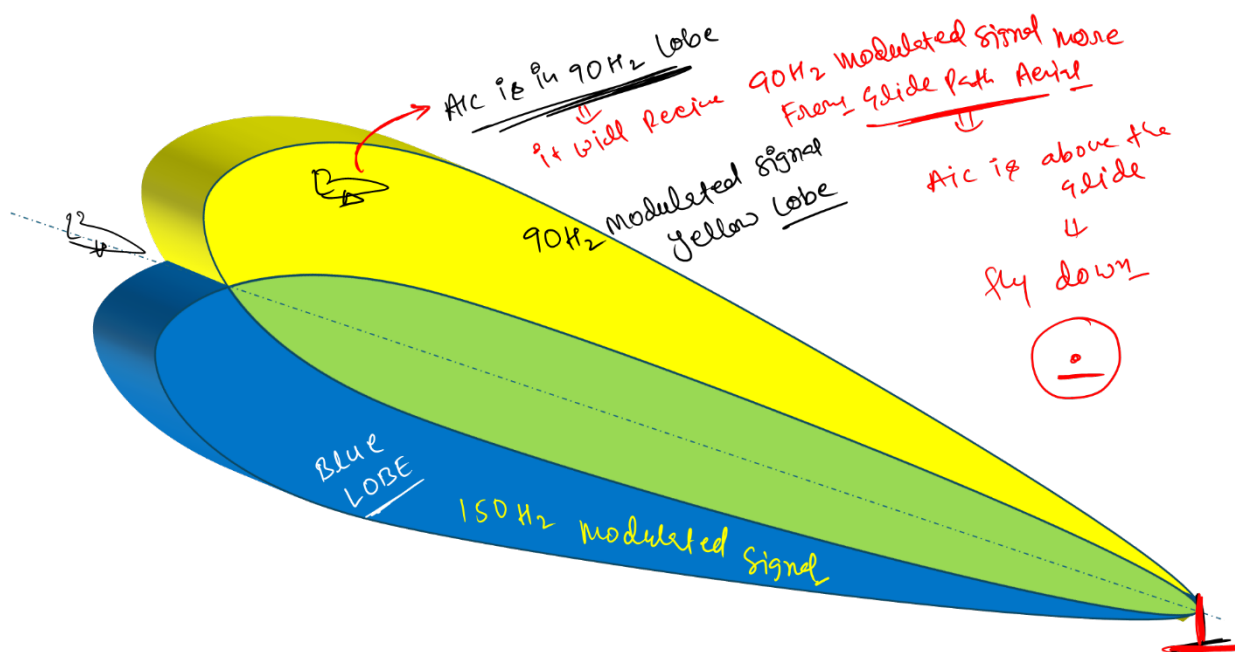


Cockpit Light	Ident	Modulating Frequency	Pitch	Touchdown Range
 BLUE	2 dashes/sec	400 Hz 	Low	6.5 – 11.1 km (3.5 – 6 NM)
 AMBER	Alternate dots and dashes 3/sec	1300 Hz 	Medium	1050 m ± 150 m (3500' ± 500')
 WHITE	6 dots/sec	3000 Hz 	High	75 - 450 m (250' – 1500')



Localizer transmits two lobes one of 90Hz modulation with yellow colour to the left and another of 150Hz modulation with blue colour to the right side.

If the aircraft is receiving 90Hz modulated signal more from the localizer that means aircraft is to the left side of centre line, so the needle in the cockpit will indicate fly right.



Like localizer the glide path aerial also transmits two lobes but 90Hz modulated to the above and 150 Hz modulated to below.

If the aircraft is receiving 90Hz modulated signal more as compared to 150Hz than aircraft is above the glide path so the needle will indicate fly down.

RADAR (Radio Detection and Ranging)

Primary Radar	Secondary Radar
One way communication	Two-way communication
Uses one antenna from transmitting and receiving the echo.	Uses transmitter and receiver
Uses echo principle for range calculation and searchlight principle to get bearing	Uses line of sight communication
Less information: Give range and bearing only	More information: apart from range and bearing it give altitude, Speed, Callsign and more

	Primary Radar	Secondary Radar
Advantage	No onboard equipment required.	- More information - Less power required
Disadvantage	- Less information - More power required	Onboard equipment required. Transponder and interrogator

Instrument using

Primary Radar	Secondary Radar
- Aerodrome surface movement Radar - Airborne weather Radar - <u>AWR</u> - Area surveillance Radar - Radio Altimeter - <u>RADALT</u>	- Secondary Surveillance Radar <u>SSR</u> - Distance measuring Equipment <u>DME</u> - Traffic collision Avoidance system <u>TCAS</u>

DME

DME stands for **Distance measuring equipment**.

It uses **secondary radar principle** and works in line-of-sight communication.

Interrogator is fitted on aircraft and the transponder is fitted on the ground.

First airborne interrogator send the signal which is received by ground base transponder and after receiving the signal it replies back after delay of microsecond on frequency ± 63 MHz.

After calculation it **displays slant range in NM**.

DME reading close to ground range when aircraft is at 100Nm or more but when aircraft is just above the station then it reads hight above the station in Nm.